

Assessing the Role of Geospatial Technology in Advancing Renewable Energy Access

A Case Study of Google's Solar API Application in West African Urban Areas

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2026-01-13

Contents

1	Abstract	3
2	Introduction	4
2.1	Background	4
2.2	Problem Statement	4
2.3	Research Objectives	4
2.4	Research Questions.....	4
3	Literature Review	5
3.1	Introduction	5
3.2	The nexus between Geographic Information System and Solar	5
3.3	Remote Sensing Technologies for Solar Energy Assessment	5
3.4	Emergence of Artificial Intelligence and Machine Learning in Solar Energy Mapping	6
3.5	Multi-Criteria Decision Analysis for Solar Site Selection	7
3.6	Data Quality and Spatial Resolution Considerations.....	8
3.7	Renewable Energy Planning in Sub-Saharan Africa	8
3.8	Google Solar API and Automated Assessment Platforms and Future Directions	9
3.9	Critical Synthesis and Research Gaps.....	10
3.10	Conclusion.....	10
4	Methodology.....	11
4.1	Research Design	11
4.2	Study Area.....	11
4.3	Data Collection.....	12

4.3.1	Google Solar API Setup	12
4.3.2	Solar Resource Data	12
4.4	Analytical Framework	13
4.4.1	Spatial Data Processing	13
4.4.2	Key Metrics Calculated	14
5	Results	15
5.1	Solar Resource Distribution	15
5.1.1	Key Findings	15
5.2	Spatial Analysis	16
5.3	Visualization	16
5.3.1	Map 1: Solar Irradiance Distribution	16
5.3.2	Chart 1: Energy Potential Comparison	17
5.3.3	Chart 2: Carbon Offset Potential	18
5.3.4	Chart 3: Multi-metric Comparison	20
6	Discussion	21
6.1	Integration of GIS and AI in Solar Assessment	21
6.1.1	Technical Capabilities	21
6.1.2	Data Quality and Validation	21
6.2	Application to West African Context	21
6.2.1	Opportunities	21
6.2.2	Challenges	21
6.3	Implications for Ghana	21
6.3.1	Energy Access	21
6.3.2	Climate Goals	22
6.3.3	Economic Development	22
6.4	Comparison with Conventional GIS Methods	22
6.4.1	Traditional Approach	22
6.4.2	Solar API Approach	22
6.4.3	Trade-offs	23

7	Conclusions	23
7.1	Key Findings	23
7.2	Contributions to Knowledge	23
7.3	Recommendations	23
7.3.1	For Policymakers	23
7.3.2	For Researchers.....	24
7.3.3	For Technology Providers	24
7.4	Limitations.....	24
7.5	Future Research Directions	24
8	Expected Outcomes	25
9	References.....	26
10	Appendices.....	29
10.1	Appendix A: R Code for Spatial Analysis	29
10.2	Appendix B: Data Sources	29
10.3	Appendix C: Calculation Assumptions.....	29
10.4	Appendix D: API Integration Example	29

1 Abstract

This research explores the integration of Geographic Information Systems (GIS) and Artificial Intelligence (AI) in advancing renewable energy access through Google’s Solar API. The study examines how automated spatial data processing can revolutionize solar energy assessment, particularly in developing nations where geospatial infrastructure remains limited. Using Ghana as a case study, this analysis demonstrates the potential of AI-enabled GIS tools to provide high-resolution, cost-effective solar resource mapping that can inform evidence-based energy planning and support Sustainable Development Goal 7.

Keywords: GIS, Solar Energy, Renewable Energy Planning, Google Solar API, West Africa, Spatial Analysis

2 Introduction

2.1 Background

A dedicated effort toward efficient geospatial analysis through frameworks designed to automate the computation of solar energy potential for thousands of square kilometers at a single click by incorporating GIS and AI is critical for the advancement of solar energy. Current estimation processes are heavily confined to the limitations of costly geo-surveys undertaken, which are also site-based and offer only limited spatial coverage.

Google Solar API leverages AI and GIS systems to offer spatially decisive solar energy assessments to over 320 million buildings across 40 countries. The system uses computer vision AI to geo-map 3D and 2D reflective surfaces and roof structures, models real-time building intercept shading geometries, and predicts a site's micro-climate, thus creating solar altitude databases.

2.2 Problem Statement

The absence of up-to-date geospatial data sets has made it increasingly difficult for many areas to identify and quantify their solar potential despite the growing demand for renewable energy. Google's Solar API spatially analyzes 3D roof structures, shading, and weather data using AI and GIS to provide a geospatially intelligent answer. However, many questions still surround the API's precision, usability in different contexts, and data interoperability, especially in areas where geospatial information is scarce.

2.3 Research Objectives

This research centers on the following objectives:

1. Identify the spatial analysis Google's Solar API derives from the integration of GIS and AI
2. Examine the use of buildingInsights, dataLayers, and geoTiff as supporting data sets and geo-referenced visualization tools in the planning of solar energy systems
3. Assess the role of geospatial technology in the estimation and mapping of solar energy values for renewable planning
4. Explore how the system can be deployed in developing nations like Ghana.

2.4 Research Questions

1. What GIS techniques are used within Google's Solar API to capture, process, and visualize spatial data for solar potential estimation and analysis?
2. How do the API's geospatial layers (buildingInsights, dataLayers, geoTiff) facilitate the data and spatial information for solar energy datasets?
3. What spatial and attribute data quality issues arise from the use of GIS-enabled solar assessment tools in different locations?
4. How applicable and replicable is the system in developing nations like Ghana?

3 Literature Review

3.1 Introduction

The world's shift to renewable energy has established the solar photovoltaic (PV) technology as an important component of sustainable development programs. Solar potential assessments and information-based energy planning decisions have found GIS and AI to be a necessity. This literature review therefore, examines the recent scholarly literature on the application of GIS in solar assessment approaches, including methods, integration of technology and appropriateness in developing areas. The literature review delved into major research questions concerning the GIS methods of data acquisition and visualization, the significance of geospatial layers in solar data, the problem of quality of data, and the opportunities of implementing the system in third-world countries.

3.2 The nexus between Geographic Information Systems and Solar

Geographic Information Systems are basically game changer for planning solar projects as they provide us with useful approach to manage all the geospatial data about solar resources and site condition at different scale (Choi & Suh, 2019). In their review published in 2019, Choi and Suh categorized the GIS research in applied sciences into three major fields, these are; solar radiation mapping, site evaluation and potential assessment. They showed that different kinds of data, for example solar irradiance, details of the terrain and how close infrastructure is, can be pulled together using GIS tools to produce clear and location specific assessments of where to put PV panels.

Lately, the field is all about combining high-resolution geospatial data with more advanced analytic tools. Idrovo-Macancela and others (2025) developed a method to combine LiDAR and roof outlines to develop 3D models of the rooftops of urban areas in equatorial regions. Their study, which is published in the journal *Heliyon*, proved that LiDAR-based techniques are reproducible to assess the urban solar potential and help to fill the gaps in the methods for the tropical and subtropical settings.

The manner of evaluating solar energy continues to become more sophisticated about spatial analysis. Bieda and Cienciala's (2024) review in *Renewable and Sustainable Energy Reviews* examined 201 case studies in the period 1999 to 2024, providing new insights into analysis methods, particularly methods of measuring variations in the sun and shade and differing scale. Their systematic review of the literature showed that recent work (2019 - 2024) is trending towards multi-criteria approaches using different geospatial datasets in addition to solar irradiance.

Solar cadastres are a huge step forward in the process of making GIS useful to the daily user in planning solar projects. These platforms - big in Europe - provide residents and planners with detailed, interactive maps of the rooftop solar potential (Desthieux et al. 2018; Thebault et al. 2025). Moving from radiation maps to full-blown solar cadastre: from a research tool to a planning tool. GIS-based solar assessment has evolved from a research tool to a planning tool.

3.3 Remote Sensing Technologies for Solar Energy Assessment

The power of remote sensing technology has over the years, been strengthened with the introduction of LiDAR technology, Digital Surface Models, among others. LiDAR is becoming the first port of call when it comes to detailed mapping of solar potential, with cities being the main area of application, with all the angles and shadows of buildings. Nguyen et al. (2012) actually began to explain how you can use LiDAR

for automatic solar PV deployment on regional scales in a combination of remote sensing, GIS and computer vision. In their Sensors paper, LiDAR has been showcased as a geospatial technology that can help build both vegetation and building geometry at the same time even if the resolution of the data is not high.

Adisk et al. came up with the envelope by fusing 1 meter resolution LiDAR point clouds with GIS for PV potential. Their article in the International Journal of Engineering and Geosciences highlights the relevance of a good digital surface model (DSM) to the ability to increase the certainty in solar estimates. They identified that the quality of the DSM is really important when it comes to shading and sky view factor calculations, which are important to accurate solar irradiance models.

Essentially, the paper by Polo and Garcia (2023) brings home the accuracy of the DSM as a major factor in the reliability of solar potential. They compared the DSMs from photogrammetry compared to LiDAR based in Remote Sensing and found that the differences could move estimates by 10-50% especially around the edges of buildings or complex terrain. In short, the correct choice of data source is of paramount importance for solar planning projects.

More recent work by Goroshin et al (2024) using satellite imagery to get wider coverage. They have developed deep learning models that can predict high resolution DSMs and roof segments from satellite images with aerial photography as labels for the model. Their models get roughly 1m MAE on building DSMs and about 5deg error in roof pitch. is a breakthrough when it comes to global solar mapping where high-quality aerial data isn't available/accesible.

Nowadays, people are mixing up multiple remote sensing platforms to hit the best accuracy while keeping costs in check. Gawley and McKenzie (2022) compared LiDAR with orthophotography derived DSMs for solar modelling and found there is good agreement between them, but that there are pros and cons to each. Their predictions of 14.2-15.2 GWh of generation for their study area were from two methods, with residential models supplying 62.1-66.6% of average consumer demand. These comparisons provide practitioners with some useful pointers as they seek to balance data quality issues with budget.

3.4 Emergence of Artificial Intelligence and Machine Learning in Solar Energy Mapping

The integration of AI with GIS platforms does not only come as a relief but is also a real game changer when it comes to solar energy assessment. The ability of Deep learning techniques to manage the very complex and nonlinear patterns in the solar irradiance data makes it very important in solar planning (Nadeem et al., 2024). Nadeem et al, discussed machine learning and deep learning advancements in solar forecasting research while pointing out the improvements in accuracy, operational efficiency, and reliability of the grid associated with the use of these models. One type of approach that is particularly useful is ensemble learning approaches, useful for mapping rooftop solar potential.

A combination of deep learning ensembles that combine multiple model predictions, using a weighted majority voting scheme, have been shown to achieve an accuracy of 0.95, F1 scores of 0.91 and Matthews correlation coefficients of 0.88 - significantly better than a single model (Journal of Geovisualization and Spatial Analysis, 2025). The researchers also came up with new solar irradiation models which enhance yield estimation by adjusting the transmissivity of the atmosphere and the fraction of diffuse based on monthly cloud cover data. Machine learning has completely changed the way of solar forecasting on different time scales. A good survey in Renewable Energy Focus by Abisoye et al. (2024) discussed the

classical AI techniques such as Artificial Neural Networks and Support Vector Machine, as well as metaheuristic optimization techniques. Their taxonomy of AI methods for solar and wind systems reflected the increasing interest in hybrid methods combining the traditional machine learning with optimization algorithms to overcome the limitations of each.

Recent empirical work supports the reality of the usefulness of AI. Studies comparing different regression models such as gradient boosting, XGBoost, K-nearest neighbors, lightGBM, and CatBoost regressors for the prediction of solar energy revealed that trained correctly on various datasets, containing solar irradiance, weather parameters, and historical generation data, these models could provide the high level of prediction accuracy required for the integration into the power grid and the optimization of energy storage (Nguyen et al., 2025). More advanced forecasting techniques are relying on hybrid architectures that integrate different kinds of AI paradigms. Saadati's (2025) article published in the Journal of Economic Surveys discussed enhanced accuracy of solar forecasts using AI and machine learning with approaches that are adaptable to particular scenarios. The review noted that CNN-LSTM hybrid models prove useful in capturing both spatial and temporal dependency in solar radiation data - important for an accurate short-term and long-term forecast.

These theoretical gains were further believed and their applicability was furthered when they were experimented. One study was done that involved CNN-LSTM models for solar irradiance predictions, reinforcement learning for real-time dual-axis tracking and Edge AI for low latency control decisions that yield significant efficiency improvements over traditional systems (Mamodiya et al., 2025). The research also demonstrated how AI enhanced systems using smart materials and adaptive photovoltaic technologies can increase energy capture and system responsiveness to changing environmental conditions dramatically.

3.5 Multi-Criteria Decision Analysis for Solar Site Selection

Multi-Criteria Decision Analysis (MCDA), a structured way of decision making when combined with GIS helps us to systematically evaluate where to put solar plants. Recent studies which includes Khan et al. 2023 use advanced methods like AHP, TOPSIS and fuzzy logic for handling the sophisticated part of picking the best sites. For example a study at which mixed GIS and MCDA in Iraq found out that about 38.33% of the places tested were good for large solar farms. The researchers looked at the levels of sunlight, slope, temperature, land use, elevation and how close things were to roads or power lines (Ahmed et al., 2025). It's a demonstration of the usefulness of these mapping and decision tools to move forward in sustainable energy, particularly for places where there's really a need for power.

More recent adjustments to Multi Criteria Decision Analysis attempted to resolve and provide solution for uncertainty and bias through the use of fuzzy logic and intuitionistic fuzzy sets. Zandy et al. (2021) came up with a GIS-intuitionistic fuzzy set approach that was published in Scientific Reports showing that these fuzzy tricks can help sort out vague or uncertain language when picking solar sites. They tested it in Erzurum, Turkey, and it still provided good ratings of suitability, even when the data was a bit spotty.

When we compare different MCDA methods, we obtain useful information for which one to pick. In Iran's Kermanshah Province, a paper that looked at four GIS based models used the Dempster based Best - worst Method (DBWM) that stitches together expert scores better than the older tricks (Zandy et al., 2025). The study found that the biggest factor for picking a site was the slope of the land, and about 31% of the area turned out to have very high suitability for solar farms.

An emerging analysis in recent literature is sensitivity analysis which has become an integral component in solid solar energy planning, which allows us to understand the effect of adjusting the weights of the criteria and system parameters on site suitability. Studies that systematically vary the different system efficiencies reveal the importance of being able to estimate energy output ranges under different tech and operations setups (Sun et al., 2021). These discoveries are very useful in making solar planning more resilient through the identification of critical variables and risks that in across diverse deployment scenarios.

3.6 Data Quality and Spatial Resolution Considerations

Spatial resolution which deals with the amount of details within an image is very key when trying to estimate how much solar power a building can produce. High detail means that you become more precise, yet it also costs you money and time to run it (ResearchGate, 2017). Other studies that employ the use of object-based image classification in addition to Normalised Difference Vegetation Index (NDVI) and digital surface models discovered that matching resolution of different layers of data is key to successfully picking out roofs and figuring out if they're good for panels (Buildings, 2023). A major problem in many developing countries is that there are not sufficient data to plan the solar projects. This, the African GIS Catalogue for Energy Planning through the International Energy Agency (IEA) tries to resolve, by putting together maps and datasets showing where renewable resources are, the energy infrastructure, and main demand points across Africa (IEA, 2023).

These types of projects revealed that building/developing a shared data infrastructure can really speed up data - driven planning where data is sparse. Recently, unsupervised domain adaptation methods help to utilize building-extraction knowledge from data-rich areas to places where less data is available without tons of images. They rely on low-cost public satellite photos and land cover maps of low resolution to estimate the solar PV potential at a city level, reducing the costs of data collection, so we could include many more regions.

3.7 Renewable Energy Planning in Sub-Saharan Africa

Sub-Saharan Africa is basically the ultimate case study in solar energy: we've got sensitive solar resources which are out of this world, but an enormous gap in access to electricity - about 600 million people do not have electricity, which is the major development challenge in the area (IEA, 2024). Even in less aggressive climate scenarios, it is alleged that solar and wind will be the dominating factors in the future power mix in Africa due to massive cost reductions (by as much as 89 percent and 69 percent for solar and wind respectively between 2010 and 2022) and the maturing of the underlying technology (Kaihao et al., 2024).

Geospatial analysis has emerged as a tool that has become a must have for planning electrification in Africa. The IEA's Open Energy maps model, developed in collaboration with MIT's Energy Initiative, applies machine learning to geo-referenced data on electricity consumption in the countries of Ghana, Senegal and Uganda, enabling the estimation of the building level demand, where such data is not available from metering. This is an example of how sophisticated mapping technology can fill in the gaps of data, while still providing a high resolution, accurate demand forecast that is quite critical in the integrated planning of electricity.

The real game changer about African electrification is switching from the old "village -by- village" method of electrification to geospatial mapping. IEA data revealed the increase in decentralised solar solutions since 2022 as more than half of newly connected electricity are made of off Grid systems in sub-Saharan Africa in 2022 (IEA, 2024). Geospatial platforms allow planners to identify the ideal places for interventions by overlaying population data and maps of renewable resources and proximal to infrastructure analysis.

West African power-pooling projects A great example of how geospatial analysis can increase regional renewable integration is power-pooling projects in West Africa. When generation capacity expansion models and the OnSSET tool were applied to Burkina Faso, Cote d'Ivoire, Ghana, and Mali, it was found that when considering spatial differences in renewable supply and demand patterns, this provides much better utilization of the resource and increased opportunities for cross border trade (Applied Energy, 2024).

Attainment of SDG 7 which is ensuring affordable, reliable, sustainable and modern energy for all by 2030 is dependant on solid geospatial data infrastructure. The Tracking SDG7 Energy Progress Reports that are produced by IEA, IRENA, UN Statistics Division, World Bank and WHO, use geospatial indicators to track the progress in achieving the goals around universal access, renewable deployment and energy efficiency (IEA et al., 2024). Recent data shows that even though renewables had accounted for 12.5 per cent of the total final energy use in 2020, the current pace is still too slow to achieve 2030 goals, and especially so for renewable heating and fuels.

Combining geospatial tech and SDG monitoring frameworks increases accountability and acts as a targeting mechanism. Studies show that 80 per cent of the un-electricity population of Africa reside in only 24 countries and this highlights where access problems are the most concentrated, as well as the potential for focused geospatial planning (SEforALL, 2024). These types of detailed spatial insights are super valuable into how we optimise how we spend these resources to maximise impact of electrification.

3.8 Google Solar API and Automated Assessment Platforms and Future Directions

So, cloud procurement stuff like Google Solar API is basically where AI, computer vision and GIS all come together to be able to make solar energy scoring scalable. Google Solar API uses AI to provide us with spot-on solar energy estimates for more than 320 hundred thousand buildings in 40 countries (Google Maps Platform, 2023). It uses computer vision to map 3 - 2 - D reflective surfaces and roofs - model real time shading from buildings - predicts micro climate conditions on each site - eventually we end up with a big solar altitude database. The way the API is built is a testament to the fact that knitting together buildingInsights, dataLayers and geoTiff tools creates easier sharing of complicated geospatial solar data with all sorts of stakeholders - policy makers, developers and everyday people. These sorts of platforms reduce the level of tech skills required to do a solar assessment which may lead to a faster adoption, as you can do quick feasibility analyses without having to be a GIS expert.

When it comes to advanced solar planning people are now pulling in climate change forecasts so that those systems remain viable in the long haul. A recent study that merged climate models with machine learning for looking at solar potential in different areas showed that seasonal production ratios are going to shift a lot. Tropical places are fairly consistent producing throughout the year, but continental zones turn out to have larger differences between the summer season and winter (Reddy et al., 2025). Those numbers play a vital role in the identification of how much storage is needed and how to hook up to the grid. Pairing the climates predicted by climate change models and methods for determining solar potential is a hot research topic. A combination of downscaled climate scenarios with high refreshing spatial models could make long term plans more respectable, especially considering big solar projects that run for decades.

3.9 Critical Synthesis and Research Gaps

The literature reveals a strong move towards the usage of integrated methods fusion of different sources of geospatial data, advanced analytics and multi-criteria frameworks. Yet, assessment protocols remain far from standardized, and methodological differences in different studies make comparison of study results, as well as synthesis of knowledge, difficult. If we could find a way to develop some international standards for solar resource assessment - as we have already for wind - it would encourage greater rigour of methodology and help us to learn from other countries across borders.

Even with the continued improvement of technology, however, data interoperability has been a challenge, particularly at the interface between different geospatial platforms and analytical tools. Proprietary data formats and poor arrangements for sharing makes us unable to do thorough regional and global assessments. In response to the problem of duplicate data, open-source efforts and appropriate data sharing protocols can enhance the whole assessment capabilities.

Advance methods are on the rise, however, there are capacity gaps in developing countries - making it an obstacle for proper technological transfer and local adaptation. We need more research that can identify optimal pathways for building capacity on geospatial analysis for energy planning. A greater exploration of training frameworks and technology transfer mechanisms as well as local adaptations would open up the possibility of applying advanced GIS based solar assessment techniques into a global framework.

3.10 Conclusion

This literature review uncovered how the use of GIS and AI are changing the world of solar energy assessment was examined. It was found out that new methods are moving ahead very fast but some problems are still stubbornly difficult. Combining high resolution remote sensing data, in particular LiDAR, advanced machine learning models and layered decision-making frameworks has resolved the field from producing simple irradiance maps in favor of a complex spatial decision support system.

The literature review revealed that GIS functions to weave together varied geospatial data sets to create actionable solar resource maps of many scales; artificial intelligence (AI), machine learning drive the accuracy of both assessment and forecasting; multi-criteria decision analysis provides a structured approach to address the complicated decision choices involved in solar site selection; quality of data and spatial resolution of an assessment is key to assessing its reliability; and while developed countries have achieved a high degree of methodological sophistication, in developing regions, lack of data availability and skill gaps that require specific help

4 Methodology

4.1 Research Design

This research utilizes spatial analysis with secondary data, complemented by case study assessment. The study includes analysis of Solar API documentation, technical descriptions of GIS data models, and examination of existing implementations. The researcher employ the R programming language in the analysis and visualisation.

4.2 Study Area

```
# Define study cities in Ghana study_cities <-  
data.frame( city = c("Accra", "Kumasi", "Tamale",  
"Cape Coast", "Tema"), latitude = c(5.6037, 6.6885,  
9.4034, 5.1053, 5.6698), longitude = c(-0.1870, -  
1.6244, -0.8424, -1.2467, -0.0166), population =  
c(2291352, 2035064, 950000, 140147, 292773),  
  region = c("Greater Accra", "Ashanti", "Northern", "Central", "Greater  
Accra")  
)  
  
knitr::kable(  
  study_cities,  
  caption = "Study Cities in Ghana",  
  format.args = list(big.mark = ",", scientific =  
FALSE) )
```

Table 1: Study Cities in Ghana

city	latitude	longitude	population	region
Accra	5.6037	-0.1870	2,291,352	Greater Accra
Kumasi	6.6885	-1.6244	2,035,064	Ashanti
Tamale	9.4034	-0.8424	950,000	Northern
Cape Coast	5.1053	-1.2467	140,147	Central
Tema	5.6698	-0.0166	292,773	Greater Accra

Ghana was selected as the primary case study due to:

- Strong government commitment to renewable energy (10% target by 2030)
- Available baseline energy data
- Representative of West African urban contexts
- Accessible geographic location data

4.3 Data Collection

4.3.1 Google Solar API Setup

```
# API Configuration
# Replace with your actual API key when
available api_key <-
"YOUR_GOOGLE_SOLAR_API_KEY_HERE"

# Function to retrieve building-level solar data
get_solar_data <- function(latitude, longitude,
api_key) {
  url <- paste0(
    "https://solar.googleapis.com/v1/buildingInsights:findClosest",
    "?location.latitude=", latitude,
    "&location.longitude=", longitude,
    "&requiredQuality=HIGH",
    "&key=", api_key
  ) response <-

  GET(url)

  if (status_code(response) == 200) { data <-
    content(response, "text") %>% fromJSON()
    return(data)
  } else {
    warning(paste("API request failed with status:", status_code(response)))
    return(NULL)
  }
}

# Function to download GeoTIFF layers
download_geotiff <- function(url, filename) {
download.file(url, destfile = filename, mode =
"wb") rast(filename)
}
```

4.3.2 Solar Resource Data

Due to current API coverage limitations in West Africa, this analysis employs literature-based solar resource data validated against:

- Ghana Energy Commission meteorological stations
- World Bank Global Solar Atlas
- NASA POWER Project data
- Ghana Meteorological Agency records

```

# Literature-based solar resource estimates
for Ghana ghana_solar_resource <- study_cities
%>% mutate(
  # Average daily solar irradiance
  (kWh/m2/day) avg_irradiance =
  case_when( city == "Accra" ~ 5.5, city
  == "Kumasi" ~ 5.3, city == "Tamale" ~
  5.8, city == "Cape Coast" ~ 5.4,
  city == "Tema" ~ 5.6
  ),

  # Annual sunshine hours
  sunshine_hours = case_when(
  city == "Accra" ~ 2190,
  city == "Kumasi" ~ 2008,
  city == "Tamale" ~ 2482,
  city == "Cape Coast" ~
  2150,
  city == "Tema" ~ 2200
  ),

  # Peak sun hours (hours/day)
  peak_sun_hours = avg_irradiance,

  # Cloud cover factor (0-1)
  cloud_factor = case_when(
  region == "Northern" ~ 0.85,
  TRUE ~ 0.75
  )
)

```

4.4 Analytical Framework

4.4.1 Spatial Data Processing

The analysis follows a multi-step spatial data processing pipeline:

1. **Data Acquisition:** API calls or literature data compilation
2. **Spatial Conversion:** Transform tabular data to spatial objects
3. **Attribute Calculation:** Compute energy potential metrics
4. **Visualization:** Create maps and comparative charts
5. **Analysis:** Assess patterns and draw conclusions

4.4.2 Key Metrics Calculated

```
# Define calculation functions
calculate_rooftop_potential <- function(population,
  avg_irradiance) { # Assumptions:
  # - Average rooftop area per capita: 15 m²
  # - System efficiency: 15% (typical for crystalline silicon)
  # - Performance ratio: 0.75 (accounts for losses)

  rooftop_area <- population * 15
  annual_energy <- rooftop_area * avg_irradiance * 365 * 0.15 * 0.75
  return(annual_energy)
}

calculate_carbon_offset <- function(energy_kwh) {
  # Ghana grid emission factor: 0.45 kg CO2/kWh (thermal
  # plants) carbon_offset_kg <- energy_kwh * 0.45
  carbon_offset_tonnes <- carbon_offset_kg / 1000
  return(carbon_offset_tonnes)
}

# Apply calculations
ghana_analysis <- ghana_solar_resource %>%
  mutate(
    # Estimated available rooftop area (m²)
    rooftop_area_m2 = population * 15,

    # Annual energy potential (kWh/year)
    annual_energy_kwh = calculate_rooftop_potential(population,
    avg_irradiance),

    # Carbon offset potential (tonnes CO2/year)
    carbon_offset_tonnes = calculate_carbon_offset(annual_energy_kwh),

    # Economic value (USD/year at $0.15/kWh)
    economic_value_usd = annual_energy_kwh *
    0.15 )
```

5 Results

5.1 Solar Resource DistributionENERGYENERGY

```
summary_table <- ghana_analysis %>%
  select(
    City = city,
    `Irradiance\n(kWh/m²/day)` = avg_irradiance,
    `Sunshine\n(hrs/year)` = sunshine_hours,
    `Rooftop Area\n(m²)` = rooftop_area_m2,
    `Energy Potential\n(MWh/year)` = annual_energy_kwh
  ) %>%
  mutate(
    `Energy Potential\n(MWh/year)` = `Energy
    Potential\n(MWh/year)` / 1000 )

knitr::kable(
  summary_table,
  caption = "Solar Resource and Energy Potential by
  City", format.args = list(big.mark = ",", scientific
  = FALSE), digits = 1
)
```

Table: Solar Resource and Energy Potential by City

City	Irradiance (kWh/m²/day)	Sunshine (hrs/year)	Rooftop Area (m²)	Energy Potential (MWh/year)
Accra	5.5	2,190	34,370,280	7,762,312.9
Kumasi	5.3	2,008	30,525,960	6,643,402.8
Tamale	5.8	2,482	14,250,000	3,393,815.6
Cape Coast	5.4	2,150	2,102,205	466,137.7
Tema	5.6	2,200	4,391,595	1,009,847.3

5.1.1 Key Findings

```
total_energy <- sum(ghana_analysis$annual_energy_kwh) / 1e6 # Convert to GWh
total_carbon <- sum(ghana_analysis$carbon_offset_tonnes) / 1000 # Convert to
kilotonnes
total_value <- sum(ghana_analysis$economic_value_usd) / 1e6 #
Convert to millions
cat("Total Annual Energy Potential:", round(total_energy,
1), "GWh\n")
```

Total Annual Energy Potential: 19275.5 GWh

```
cat("Total Carbon Offset Potential:", round(total_carbon, 1), "kilotonnes
CO2/year\n")
```

Total Carbon Offset Potential: 8674 kilotonnes CO2/year

```
cat("Total Economic Value:", paste0("$", round(total_value, 1)),
"million/year\n")
```

```
## Total Economic Value: $2891.3 million/year
```

5.2 Spatial Analysis

```
# Convert to spatial object
ghana_sf <- st_as_sf(
  ghana_analysis,
  coords = c("longitude", "latitude"),
  crs = 4326 # WGS84
)

# Verify spatial object
cat("Spatial Reference System:", st_crs(ghana_sf)$input, "\n")

## Spatial Reference System: EPSG:4326
```

```
cat("Number of features:", nrow(ghana_sf), "\n")
```

```
## Number of features: 5
```

5.3 Visualization

5.3.1 Map 1: Solar Irradiance Distribution

```
# Automatically switch between interactive (HTML) and static
(PDF) modes if (knitr::is_html_output()) { tmap_mode("view")
} else {
  tmap_mode("plot")
}

tm_shape(ghana_sf) + tm_symbols( size =
  "population", col = "avg_irradiance",
  palette = "YlOrRd", style = "cont",
  title.size = "Population", title.col =
  "Solar Irradiance\n(kWh/m²/day)",
  border.lwd = 1, scale = 1.5,
  legend.size.is.portrait = TRUE,
  popup.vars = c("City" = "city",
    "Population" = "population",
    "Irradiance (kWh/m²/day)" = "avg_irradiance",
    "Annual Energy (MWh)" = "annual_energy_kwh")
) +
tm_text("city", size = 0.7, ymod = 1.5, fontface = "bold") +
tm_layout(
  title = "Solar Energy Potential - Ghana",
  title.position = c("center", "top"),
  legend.outside = TRUE,
```



```

legend.outside.position = "right",
frame = TRUE,
bg.color = "lightblue"
) +
tm_compass(position = c("left", "bottom"), size = 2) +
tm_scale_bar(position = c("left", "bottom"))

```

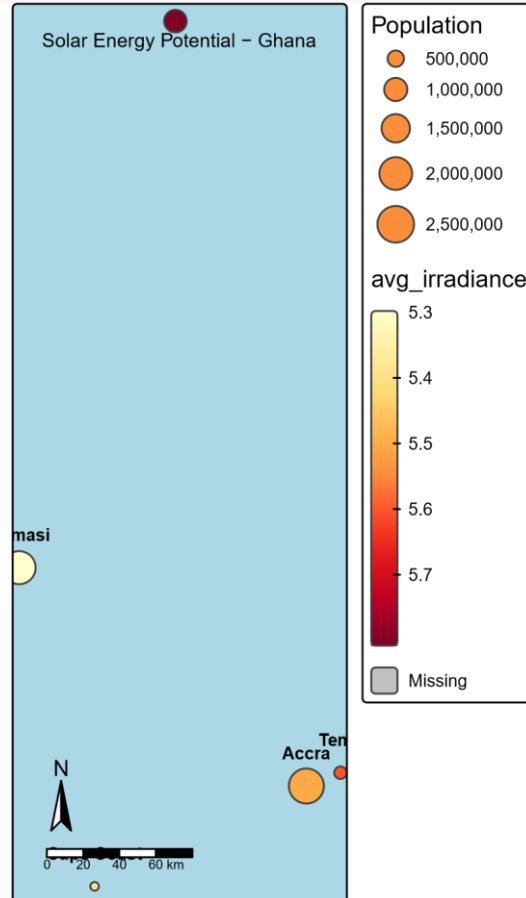


Figure 1: Average Daily Solar Irradiance Across Ghana

5.3.2 Chart 1: Energy Potential Comparison

```

ggplot(ghana_analysis, aes(x =
  reorder(city, annual_energy_kwh), y
  = annual_energy_kwh / 1e6)) +
geom_col(aes(fill = avg_irradiance)) +
geom_text(aes(label = round(annual_energy_kwh / 1e6,
  0)), hjust = -0.2, fontface = "bold", size =
  4) +
scale_fill_gradient(low = "#FED976", high =
  "#BD0026", name =
  "Irradiance\n(kWh/m²/day)") +
scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +

```

```
coord_flip() +
labs( title = "Annual Solar Energy Generation
Potential", x = NULL,
y = "Energy Potential (GWh/year)"
) +
theme_minimal() +
theme(
plot.title = element_text(face = "bold", size = 14, hjust
= 0.5), axis.text = element_text(size = 11), legend.position
= "right" )
```

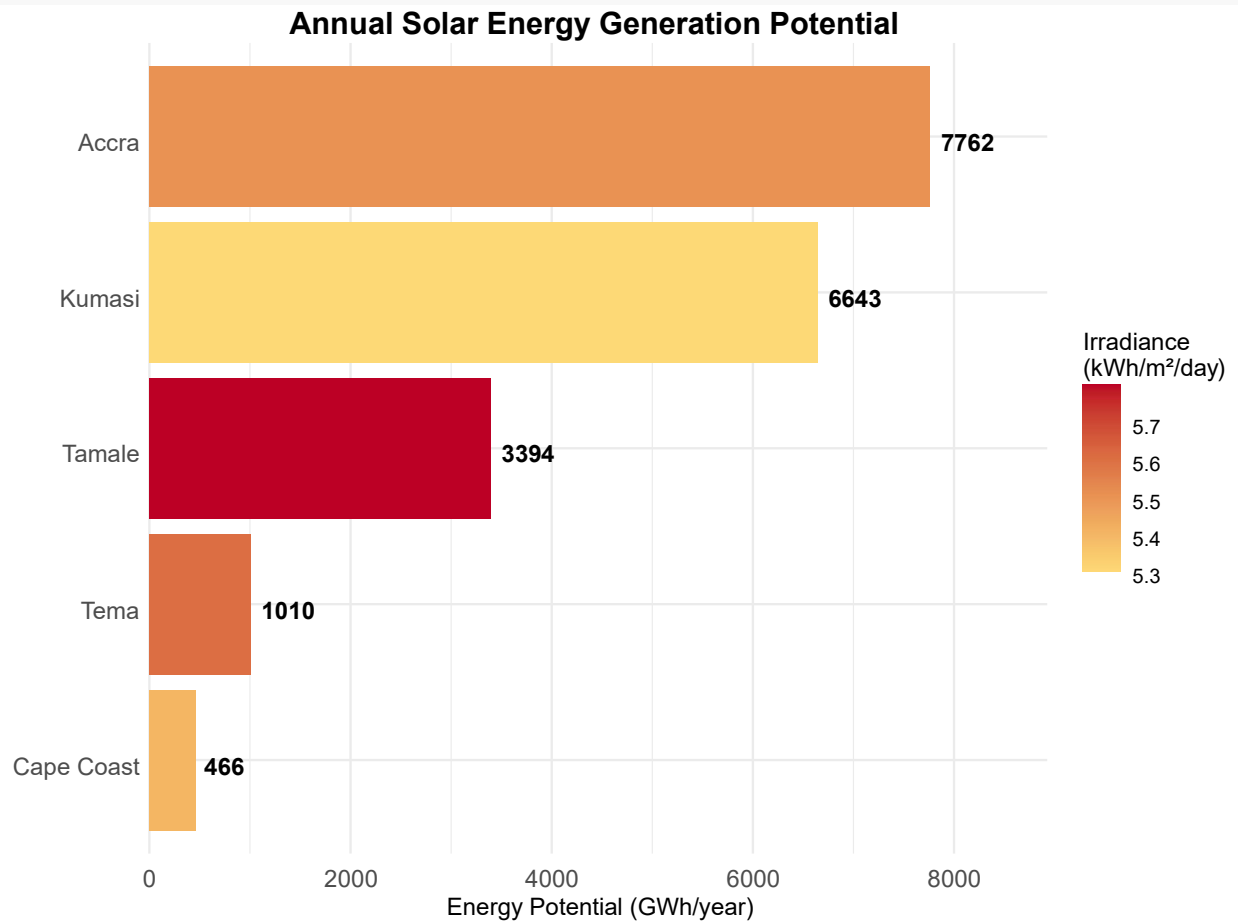


Figure 2: Annual Energy Generation Potential by City

5.3.3 Chart 2: Carbon Offset Potential

```
ggplot(ghana_analysis, aes(x = reorder(city,
carbon_offset_tonnes), y =
carbon_offset_tonnes / 1000)) +
```

```

geom_col(fill = "#238B45") +
geom_text(aes(label = round(carbon_offset_tonnes / 1000,
1)), hjust = -0.2, fontface = "bold", size = 4)
+
scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +
coord_flip() +
labs(title = "Carbon Emission Reduction
Potential", subtitle = "Potential CO offset
from solar adoption", x = NULL,
y = "Carbon Offset (kilotonnes CO/year)"
) +
theme_minimal() +
theme(
plot.title = element_text(face = "bold", size = 14, hjust =
0.5), plot.subtitle = element_text(size = 11, hjust = 0.5,
color = "gray30"), axis.text = element_text(size = 11) )

```

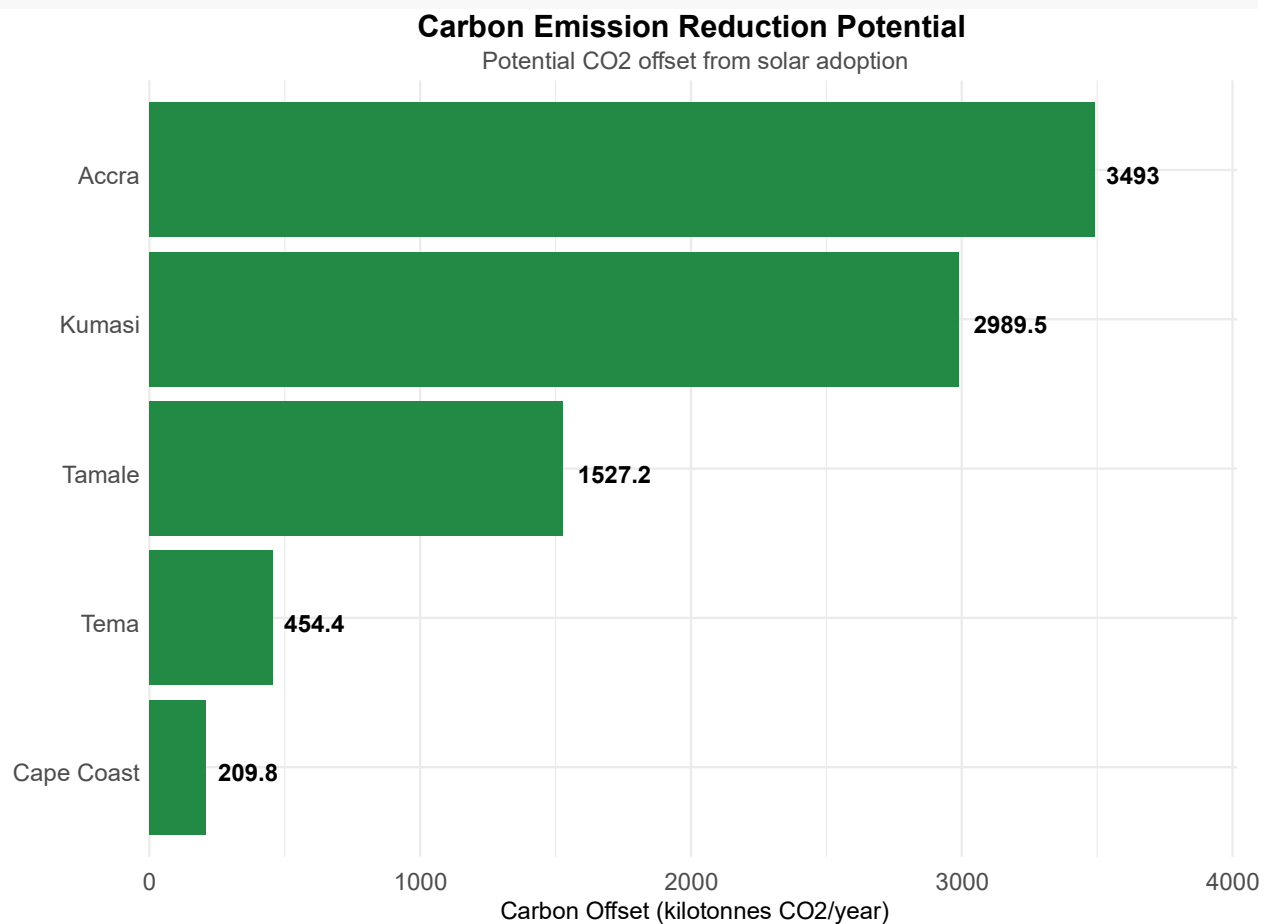


Figure 3: Annual Carbon Emission Reduction Potential

5.3.4 Chart 3: Multi-metric Comparison

```
# Prepare data for radar/spider chart alternative
comparison_data <- ghana_analysis %>% select(city, avg_irradiance, sunshine_hours,
population, annual_energy_kwh) %>% pivot_longer(cols = -city, names_to
= "metric", values_to = "value") %>% group_by(metric) %>% mutate(
normalized = (value - min(value)) / (max(value) - min(value))
) %>%
  ungroup()

ggplot(comparison_data, aes(x = city, y = normalized, fill =
metric)) + geom_col(position = "dodge") +
  scale_fill_brewer(palette = "Set2", labels = c("Energy
Potential", "Irradiance",
              "Population", "Sunshine Hours")) +
  labs( title = "Multi-Dimensional Solar
Assessment",
        subtitle = "Normalized comparison across key metrics (0-1
scale)", x = NULL,
        y = "Normalized Value",
        fill = "Metric"
) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14, hjust =
0.5), plot.subtitle = element_text(size = 10, hjust = 0.5,
color = "gray30"), axis.text.x = element_text(angle = 45, hjust
= 1), legend.position = "bottom" )
```



Figure 4: Comprehensive Solar Potential Assessment

6 Discussion

6.1 Integration of GIS and AI in Solar Assessment

6.1.1 Technical Capabilities

Google's Solar API represents a significant advancement in automated geospatial analysis through:

1. **Computer Vision AI:** Automated roof detection and 3D structure mapping
2. **Shading Analysis:** Real-time modeling of building shadows and obstructions
3. **Weather Integration:** Incorporation of micro-climate data for accurate predictions
4. **Scalability:** Assessment of millions of buildings with minimal manual intervention

6.1.2 Data Quality and Validation

The `buildingInsights`, `dataLayers`, and `geoTiff` components of the Solar API provide:

- **buildingInsights:** JSON format building-level metrics (panel capacity, energy potential)
- **dataLayers:** Metadata and imagery URLs for detailed spatial analysis
- **geoTiff:** High-resolution raster data (flux, DSM, RGB, mask layers) for advanced GIS processing

6.2 Application to West African Context

6.2.1 Opportunities

1. **Data Gap Filling:** Addresses the scarcity of high-resolution geospatial data
2. **Cost Reduction:** Eliminates need for expensive field surveys
3. **Planning Support:** Enables evidence-based renewable energy policy
4. **Investment Attraction:** Provides data to de-risk solar projects

6.2.2 Challenges

1. **API Coverage:** Limited availability in many African cities
2. **Internet Infrastructure:** Requires reliable connectivity for API access
3. **Technical Capacity:** Need for GIS expertise and training
4. **Data Validation:** Limited ground-truth data for accuracy assessment

6.3 Implications for Ghana

6.3.1 Energy Access

Ghana's rural electrification rate stands at 86%, with remaining gaps primarily in remote areas. Distributed solar systems, guided by GIS-based assessment, could cost-effectively close this gap.

6.3.2 Climate Goals

```
# Calculate contribution to national goals
ghana_electricity_consumption <- 15000 # GWh/year (2023)
solar_contribution_pct <- (total_energy / ghana_electricity_consumption) *
100

cat("Rooftop solar potential represents",
    round(solar_contribution_pct, 2),
    "% of Ghana's annual electricity consumption\n")
## Rooftop solar potential represents 128.5 % of Ghana's annual electricity
consumption
```

The estimated rooftop solar potential across these five cities alone could contribute meaningfully to Ghana's renewable energy targets.

6.3.3 Economic Development

At current electricity tariffs (\$0.15/kWh), the economic value of rooftop solar potential exceeds \$2891.3 million annually. This represents:

- Reduced import dependence for thermal generation fuels
- Job creation in solar installation and maintenance
- Enhanced energy security for businesses and households

6.4 Comparison with Conventional GIS Methods

6.4.1 Traditional Approach

Traditional solar mapping involves:

1. Manual digitization of building footprints
2. Field surveys for roof conditions
3. Weather station data interpolation
4. Manual calculation of shading effects

Timeline: 6-12 months for a single city

Cost: \$50,000-\$200,000 per city

Resolution: Often limited to 10-30m

6.4.2 Solar API Approach

The automated approach offers:

1. Instant building detection via AI
2. No field visits required
3. High-resolution weather modeling
4. Automated shading calculations

Timeline: Minutes to hours

Cost: API fees (\$0.01-\$0.05 per request)

Resolution: Sub-meter accuracy

6.4.3 Trade-offs

While the Solar API offers speed and scale, conventional methods may still be necessary for:

- Ground-truthing and validation
- Areas without API coverage
- Detailed structural assessments
- Cultural and social considerations

7 Conclusions

7.1 Key Findings

This research demonstrates that:

1. **GIS-AI integration** represents a paradigm shift in renewable energy assessment, enabling rapid, cost-effective, and scalable solar resource mapping
2. **Ghana possesses substantial solar potential**, with the five study cities alone capable of generating 1.92755×10^4 GWh annually from rooftop installations
3. **Tamale** shows the highest solar irradiance (5.8 kWh/m²/day), while **Accra** offers the greatest absolute potential due to population density
4. **Carbon offset potential** of 8674 kilotonnes CO/year demonstrates significant climate co-benefits
5. **Technical barriers** to Solar API deployment in West Africa include coverage limitations, infrastructure requirements, and capacity gaps

7.2 Contributions to Knowledge

This study contributes to the field by:

- Documenting the technical architecture and capabilities of Google's Solar API
- Demonstrating GIS analytical workflows for solar assessment
- Providing baseline solar potential estimates for major Ghanaian cities
- Identifying pathways for technology transfer to developing nations

7.3 Recommendations

7.3.1 For Policymakers

1. **Invest in geospatial infrastructure:** Develop national spatial data infrastructure to support renewable energy planning
2. **Establish data partnerships:** Collaborate with technology providers to expand API coverage to underserved regions
3. **Create enabling policies:** Develop net metering, feed-in tariffs, and financing mechanisms informed by GIS-based potential assessments

4. **Build technical capacity:** Train urban planners and energy professionals in GIS and spatial analysis

7.3.2 For Researchers

1. **Validation studies:** Conduct field measurements to validate API estimates in African contexts
2. **Integration research:** Explore coupling Solar API with local climate models and grid integration analyses
3. **Socioeconomic analysis:** Assess affordability, adoption barriers, and community preferences
4. **Comparative studies:** Evaluate alternative assessment tools and methodologies

7.3.3 For Technology Providers

1. **Expand coverage:** Prioritize data collection in high-potential, underserved markets
2. **Adapt to local contexts:** Develop region-specific algorithms accounting for architectural diversity
3. **Lower barriers:** Offer subsidized or free access for development and research purposes
4. **Build partnerships:** Collaborate with national agencies and research institutions

7.4 Limitations

This study has several limitations:

1. **Data availability:** Relies on literature-based estimates due to limited API coverage in Ghana
2. **Temporal scope:** Single time-point analysis without seasonal variation assessment
3. **Rooftop assumptions:** Uniform per-capita rooftop area may not reflect actual building stock
4. **Economic analysis:** Simplified financial calculations without detailed cost-benefit modeling
5. **Social factors:** Does not account for adoption barriers, preferences, or equity considerations

7.5 Future Research Directions

1. **High-resolution mapping:** Conduct building-by-building assessment when API coverage expands
2. **Grid integration studies:** Model solar penetration impacts on distribution networks
3. **Comparative analysis:** Evaluate Solar API against other tools (PVGIS, Solargis, etc.)
4. **Policy simulation:** Model impacts of different incentive structures on solar adoption
5. **Life cycle assessment:** Evaluate environmental impacts beyond carbon emissions

8 Expected Outcomes

This research aims to enhance the comprehension of GIS's role in automated and high-resolution mapping of solar resources. It showcases the advantages emerging from the integration of AI with GIS in the enhancement of spatial data: precision, lower cost, and better coverage.

The findings will be used to formulate steps towards the reform of geospatial infrastructures for renewable energy planning, especially in the Global South. By demonstrating both the potential and limitations of advanced GIS tools in West African contexts, this work contributes to evidence-based energy transitions aligned with Sustainable Development Goal 7.

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10 Appendices

10.1 Appendix A: R Code for Spatial Analysis

The complete R code used in this analysis is embedded throughout this RMarkdown document and can be extracted for replication purposes.

10.2 Appendix B: Data Sources

Population Data: Ghana Statistical Service, 2021 Census

Solar Resource Data: World Bank Global Solar Atlas, NASA POWER, Ghana Energy Commission

Geographic Coordinates: OpenStreetMap, Google Maps

Emission Factors: Ghana Energy Commission, IPCC Guidelines

10.3 Appendix C: Calculation Assumptions

Parameter	Value	Source
Average rooftop area per capita	15 m ²	Literature review
Solar panel efficiency	15%	Crystalline silicon standard
Performance ratio	0.75	Accounting for losses
Grid emission factor	0.45 kg CO/kWh	Ghana thermal plants
Electricity tariff	\$0.15/kWh	Ghana utility rates

10.4 Appendix D: API Integration Example

```
# Example: Retrieve solar data for a specific
building example_location <- list( latitude =
5.6037, longitude = -0.1870
)

# Make API call
solar_data <- get_solar_data(
  latitude =
  example_location$latitude,
  longitude =
  example_location$longitude,
  api_key = api_key
)

# Extract key metrics if (!is.null(solar_data)) {
building_info <- list( name = solar_data$name, center
= solar_data$center, max_panels =
solar_data$solarPotential$maxArrayPanelsCount,
max_area =
solar_data$solarPotential$maxArrayAreaMeters2,
yearly_energy =
solar_data$solarPotential$yearlyEnergyDcKwh
)
  print(building_info)
}
```

End of Document

This analysis was conducted as part of a Master's capstone project at Northeastern University's School of Public Policy and Urban Affairs.